

PROJECT 5

TELLCO
TELECOM USER
ANALYTICS

~ BY SIDDHEYA LOHAR

BUSINESS PROBLEM

01

BUSINESS PROBLEM

TellCo wants to understand its customers better before making strategic decisions regarding customer acquisition and business growth.

THE ANALYSIS FOCUSES ON

- TellCo wants to understand its customers better before making strategic decisions regarding customer acquisition and business growth. Understanding who the users are
- Measuring customer engagement
- Evaluating network experience
- Measuring customer satisfaction
- Identifying high-value customer segments
- Providing insights that can support future business decisions

OBJECTIVES

FOUR CORE OBJECTIVES

USER OVERVIEW

- Number of users
- Session behaviour
- Handset manufacturers
- Handset types
- Traffic characteristics

ENGAGEMENT ANALYSIS

- Session frequency
- Session duration
- Data consumption
- Engagement score
- Customer segments

EXPERIENCE ANALYSIS

- TCP retransmission
- RTT
- Throughput
- Network experience
- Experience-based customer segments

SATISFACTION ANALYSIS

- Engagement
- Experience
- Satisfaction score
- Customer segmentation

Dataset Overview

Dataset

Metric		 
Initial rows		150,001
Initial columns		55
Duplicate rows		10
Unique customers		106,856
Sessions per customer		1-18
Final cleaned dataset		148,506 × 55

Data Dictionary

A SECOND DATASET WAS PROVIDED CONTAINING FIELD DESCRIPTIONS. IT WAS USED TO UNDERSTAND THE MEANING OF VARIABLES RATHER THAN BEING DIRECTLY MERGED WITH THE MAIN DATASET.

```

Number', 'IMSI', 'IMEI', 'Handset Manufacturer', 'Handset Type']].isna().sum())
1066
570
572
urer 572
572
572

# Numeric columns
numeric_columns = df1.select_dtypes(include="number").columns

len(numeric_columns)
50

start ms', 'End ms', 'Dur. (ms)', 'IMSI', 'MSISDN/Number', 'IMEI', 'Avg RTT DL (ms)', 'Avg RTT UL (ms)', 'Avg Bearer TP DL (kbps)', 'Avg B
s)', 'TCP DL Retrans. Vol (Bytes)', 'TCP UL Retrans. Vol (Bytes)', 'DL TP < 50 Kbps (%)', '50 Kbps < DL TP < 250 Kbps (%)', '250 Kbps < D
', 'DL TP > 1 Mbps (%)', 'UL TP < 10 Kbps (%)', '10 Kbps < UL TP < 50 Kbps (%)', '50 Kbps < UL TP < 300 Kbps (%)', 'UL TP > 300 Kbps
Bytes)', 'HTTP UL (Bytes)', 'Activity Duration DL (ms)', 'Activity Duration UL (ms)', 'Dur. (ms).1', 'Nb of sec with 125000B < Vol DL',
1250B < Vol UL < 6250B', 'Nb of sec with 31250B < Vol DL < 125000B', 'Nb of sec with 37500B < Vol UL', 'Nb of sec with 6250B < Vol DL < 3
ec with 6250B < Vol UL < 37500B', 'Nb of sec with Vol DL < 6250B', 'Nb of sec with Vol UL < 1250B', 'Social Media DL (Bytes)', 'Social Me
Google DL (Bytes)', 'Google UL (Bytes)', 'Email DL (Bytes)', 'Email UL (Bytes)', 'Youtube DL (Bytes)', 'Youtube UL (Bytes)', 'Netflix D
Flix UL (Bytes)', 'Gaming DL (Bytes)', 'Gaming UL (Bytes)', 'Other DL (Bytes)', 'Other UL (Bytes)', 'Total UL (Bytes)', 'Total DL (Byte

# Object columns
object_columns = df1.select_dtypes(include="object").columns

len(object_columns)
5

['Last Location Name', 'Handset Manufacturer', 'Handset Type']

# Categorical columns
categorical_columns = [
    'er',

```

DATA PREPARATION

- Inspected dataset structure and data types
- Removed duplicate records
- Identified missing values
- Investigated numerical distributions
- Identified potential outliers
- Reviewed categorical variables
- Examined highly correlated/redundant variables
- Prepared data for customer-level aggregation

CLEANING & PREPARING

FINAL DATASET

148,506 rows × 55 columns

MISSING VALUE

MISSING VALUES WERE EXAMINED ACROSS ALL NUMERICAL AND CATEGORICAL VARIABLES.

```
missing = df_clean.isna().sum()

missing = missing[missing > 0].sort_values(ascending=False)

print(missing)
```

Nb of sec with 37500B < Vol UL	129329
Nb of sec with 6250B < Vol UL < 37500B	111106
Nb of sec with 125000B < Vol DL	96984
TCP UL Retrans. Vol (Bytes)	96192
Nb of sec with 31250B < Vol DL < 125000B	93106
Nb of sec with 1250B < Vol UL < 6250B	92426
Nb of sec with 6250B < Vol DL < 31250B	87911
TCP DL Retrans. Vol (Bytes)	87796
HTTP UL (Bytes)	81508
HTTP DL (Bytes)	81171
Avg RTT DL (ms)	27644
Avg RTT UL (ms)	27625
Nb of sec with Vol UL < 1250B	768
10 Kbps < UL TP < 50 Kbps (%)	767
UL TP < 10 Kbps (%)	767
UL TP > 300 Kbps (%)	767
50 Kbps < UL TP < 300 Kbps (%)	767
Nb of sec with Vol DL < 6250B	721
50 Kbps < DL TP < 250 Kbps (%)	720
DL TP < 50 Kbps (%)	720
DL TP > 1 Mbps (%)	720
250 Kbps < DL TP < 1 Mbps (%)	720
Last Location Name	589
Bearer Id	429

dtype: int64

34 COLUMNS:
LESS THAN 1%
MISSING

Dur. (ms)	Total Duration of the xDR (in ms)
Start	Start time of the xDR (first frame timestamp)
Start ms	Milliseconds offset of start time for the xDR (first frame timestamp)
End	End time of the xDR (last frame timestamp)
End ms	Milliseconds offset of end time of the xDR (last frame timestamp)
Dur. (s)	Total Duration of the xDR (in s)
IMSI	International Mobile Subscriber Identity
MSISDN/Number	MS International PSTN/ISDN Number of mobile - customer number
IMEI	International Mobile Equipment Identity
Last Location Name	User location call name (2G/3G/4G) at the end of the bearer
Avg RTT DL (ms)	Average Round Trip Time measurement Downlink direction (msecond)
Avg RTT UL (ms)	Average Round Trip Time measurement Uplink direction (msecond)
Avg Bearer TP DL (kbps)	Average Bearer Throughput for Downlink (kbps) - based on BDR duration
Avg Bearer TP UL (kbps)	Average Bearer Throughput for uplink (kbps) - based on BDR duration
TCP DL Retrans. Vol (Bytes)	TCP volume of Downlink packets detected as retransmitted (bytes)
TCP UL Retrans. Vol (Bytes)	TCP volume of Uplink packets detected as retransmitted (bytes)
DL TP < 50 Kbps (%)	Duration ratio when Bearer Downlink Throughput < ...
50 Kbps < DL TP < 250 Kbps (%)	Duration ratio when Bearer Downlink Throughput range is ...
250 Kbps < DL TP < 1 Mbps (%)	Duration ratio when Bearer Downlink Throughput range is ...
DL TP > 1 Mbps (%)	Duration ratio when Bearer Downlink Throughput > ...
UL TP < 10 Kbps (%)	Duration ratio when Bearer Uplink Throughput < ...
10 Kbps < UL TP < 50 Kbps (%)	Duration ratio when Bearer Uplink Throughput range is ...
50 Kbps < UL TP < 300 Kbps (%)	Duration ratio when Bearer Uplink Throughput range is ...
UL TP > 300 Kbps (%)	Duration ratio when Bearer Uplink Throughput > ...
HTTP DL (Bytes)	HTTP data volume (in Bytes) received by the MS during this session
HTTP UL (Bytes)	HTTP data volume (in Bytes) sent by the MS during this session
Activity Duration DL (ms)	Activity Duration for downlink (ms) - excluding periods of inactivity > 500 ms
Activity Duration UL (ms)	Activity Duration for uplink (ms) - excluding periods of inactivity > 500 ms
Dur. (ms).1	Total Duration of the xDR (in ms)
Handset Manufacturer	Handset manufacturer

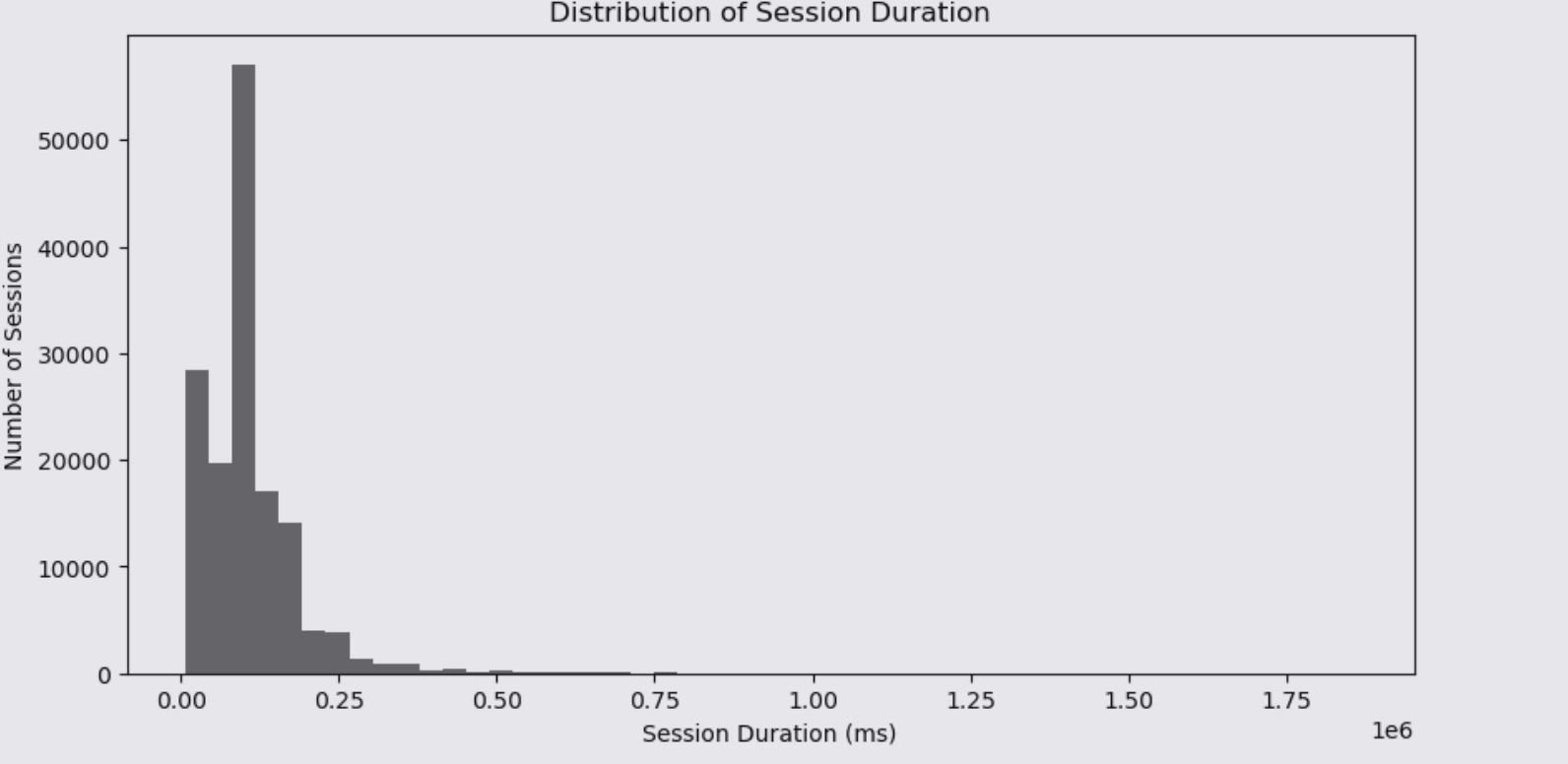
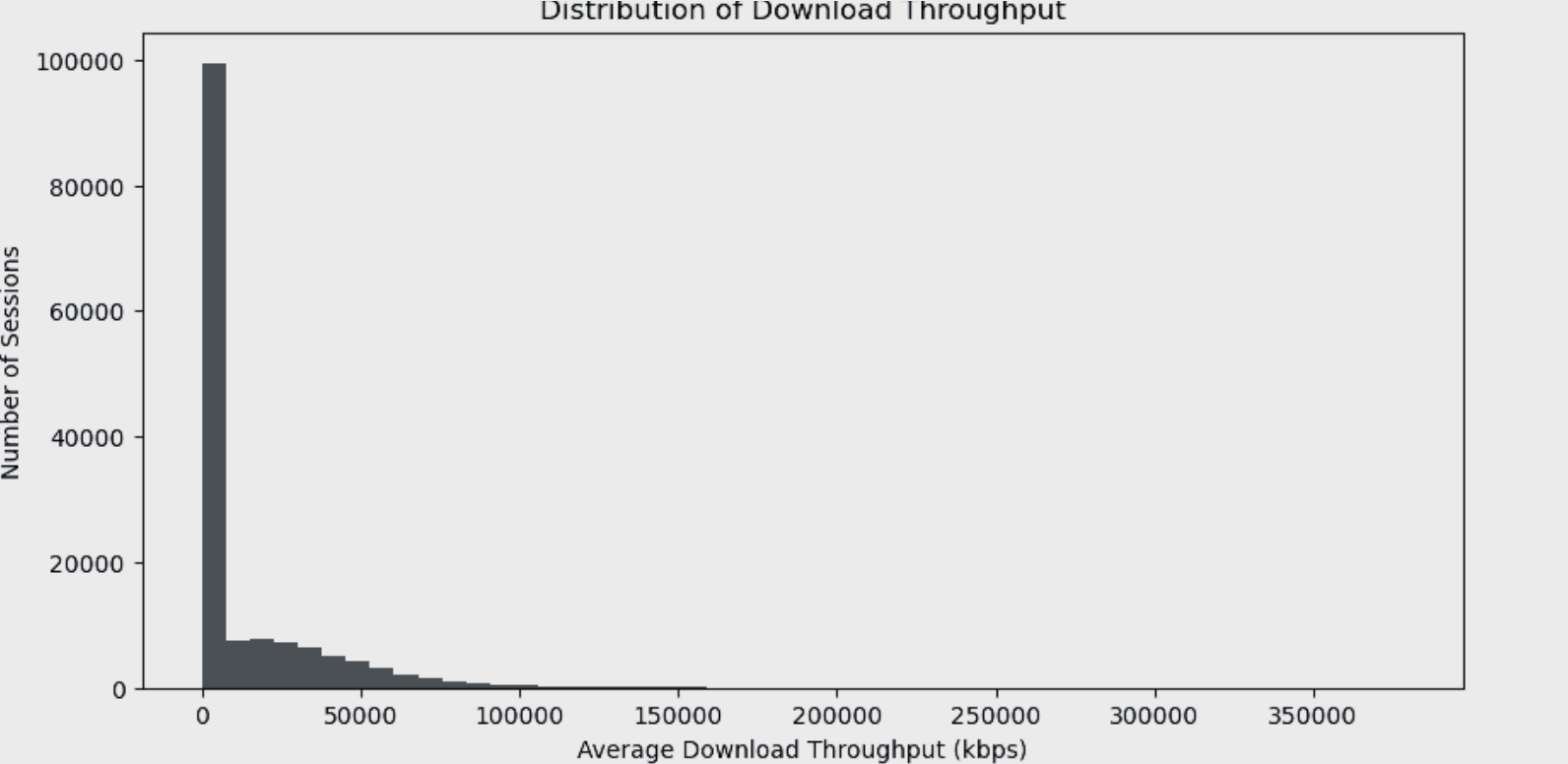
2 COLUMNS: 1–20%
MISSING

10 COLUMNS: MORE THAN
50% MISSING

IMPORTANT CORRELATION

- **DUR. VS DUR.1 → 1.000**
- **TOTAL DL VS GAMING DL → 0.999**
- **ACTIVITY DURATION UL VS DL → 0.905**
- **LOW-VOLUME UL/DL SECONDS → 0.947**

CORRELATION & REDUNDANCY ANALYSIS

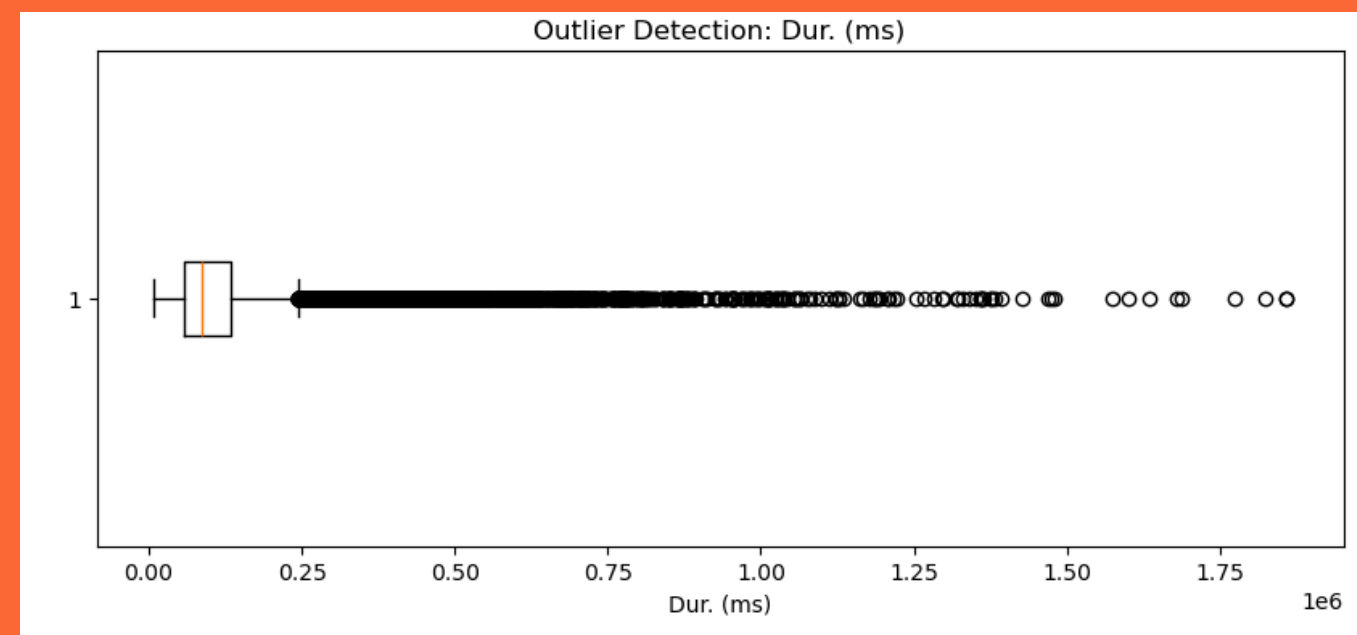


OUTLIER ANALYSIS

OUTLIER DETECTION

**DL TP > 1
Mbps (%)**

- 36,744 observations
- 24.79%



**10 Kbps < UL TP
< 50 Kbps (%)**

- 31,691 observations
- 21.39%

Most Engaged Users:

	MSISDN/Number	Number_of_Sessions	Total_Duration_ms	Total_Traffic_Bytes	Sessions_Score	Duration_Score	Traffic_Score	Engagement_Score
13180	3.362578e+10	17	18553754.0	8.499621e+09	0.941176	1.000000	0.960671	0.967282
6437	3.361489e+10	17	9966898.0	8.846226e+09	0.941176	0.537012	1.000000	0.826063
13526	3.362632e+10	18	8791927.0	7.971167e+09	1.000000	0.473660	0.900708	0.791456
92923	3.376054e+10	15	9279434.0	8.514774e+09	0.823529	0.499945	0.962390	0.761955
76363	3.367588e+10	15	4865947.0	7.891111e+09	0.823529	0.261978	0.891624	0.659044
37052	3.365973e+10	16	4035428.0	7.705863e+09	0.882353	0.217198	0.870604	0.656718
65118	3.366716e+10	13	8744914.0	5.618394e+09	0.705882	0.471125	0.633741	0.603583
92577	3.376041e+10	12	5321667.0	7.132371e+09	0.647059	0.286550	0.805530	0.579713
63028	3.366646e+10	11	4536757.0	7.308501e+09	0.588235	0.244229	0.825516	0.552660
86455	3.369879e+10	11	5169128.0	6.540899e+09	0.588235	0.278325	0.738417	0.534992

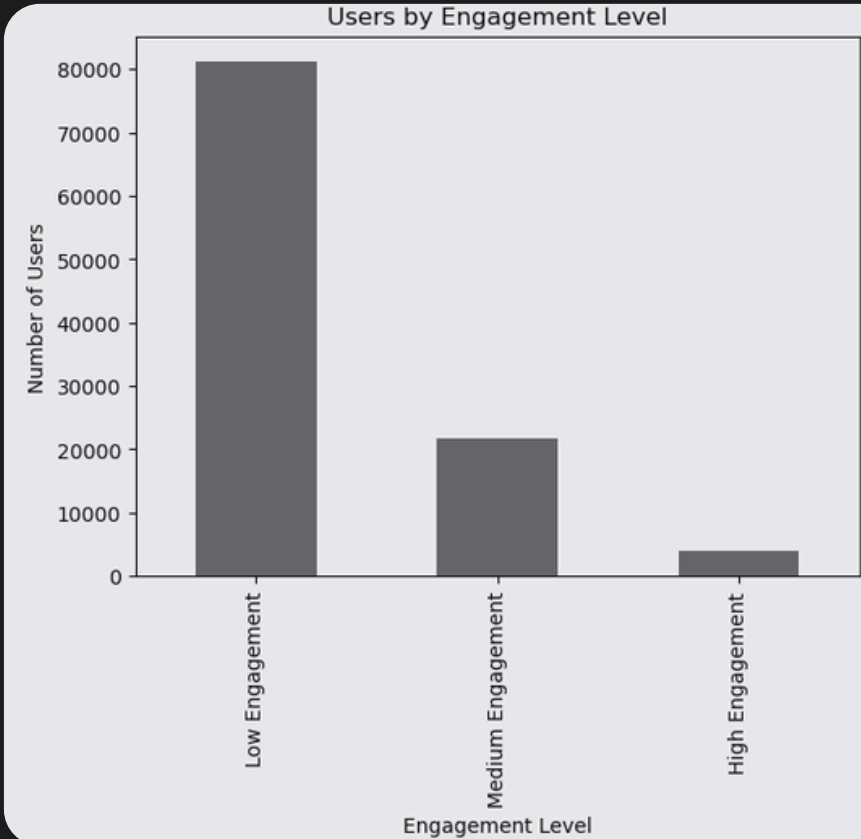
USER OVERVIEW ANALYSIS

CUSTOMER & USAGE OVERVIEW

- Number of sessions
- Session duration
- Download traffic
- Upload traffic
- Handset manufacturer
- Handset type

KEY FINDING

106,856 unique customers



SESSION FREQUENCY

How often a customer uses the service.

SESSION DURATION

How long the customer remains active.

DATA VOLUME

How much data the customer consumes.

CUSTOMER ENGAGEMENT

ENGAGEMENT SCORE & TOP CUSTOMERS

ENGAGEMENT RANKING

	MSISDN/Number	Engagement_Score	Experience_Score	Satisfaction_Score
0	3.360100e+10	0.033947	0.492868	0.263407
1	3.360100e+10	0.007804	0.494178	0.250991
2	3.360100e+10	0.023581	0.001477	INTERPRETATION • Session activity • Duration • Data consumption
3	3.360101e+10	0.015484	0.572995	
4	3.360101e+10	0.074012	0.756406	

HIGHER ENGAGEMENT SCORES INDICATE CUSTOMERS WITH STRONGER COMBINED USAGE ACROSS:

CUSTOMER SEGMENTATION

FEATURES

- SESSION FREQUENCY
- SESSION DURATION
- DATA USAGE
- ENGAGEMENT SCORE / NORMALIZED ENGAGEMENT FEATURES

K-MEANS CLUSTERING

K-MEANS CLUSTERING WAS USED TO GROUP CUSTOMERS BASED ON THEIR ENGAGEMENT BEHAVIOUR.

```
experience_kmeans = KMeans(  
    n_clusters=3,  
    random_state=42,  
    n_init=10  
)  
  
experience_clustering["Experience_Cluster"] = (  
    experience_kmeans.fit_predict(experience_clustering)  
)
```

```
experience_clustering["Experience_Cluster"] = (  
    experience_kmeans.fit_predict(experience_clustering)  
)
```

```
experience_clustering["Experience_Cluster"].value_counts().sort_index()
```

```
Experience_Cluster  
0    23684  
1     11189  
2       4488  
Name: count, dtype: int64
```

```
experience_cluster_summary = (  
    experience_clustering  
    .groupby("Experience_Cluster")  
    [  
        [  
            "Session_Frequency",  
            "Session_Duration",  
            "Data_Usage",  
            "Engagement_Score",  
            "Normalized_Engagement_Features"  
        ]  
    ]  
)
```



TCP RETRANSMISSION

Higher retransmission can indicate network inefficiency or packet delivery problems.



RTT-ROUND TRIP TIME

Measures communication latency between the user and network.



THROUGHPUT

Measures the effective data transfer rate experienced by users.

NETWORK EXPERIENCE ANALYSIS

	Avg_Throughput_DL	Avg_Throughput_UL
count	1394.000000	1394.000000
mean	11239.881815	1051.076311
std	13637.699937	1977.274199
min	0.000000	0.000000
25%	1053.500000	180.000000
50%	7630.014860	561.000000
75%	15845.375000	1123.034051
max	104669.000000	27277.250000

EXPERIENCE SCORING & SEGMENTATION

FOR METRICS SUCH AS TCP RETRANSMISSION AND RTT, LOWER VALUES GENERALLY REPRESENT BETTER NETWORK CONDITIONS, WHILE HIGHER THROUGHPUT REPRESENTS BETTER PERFORMANCE.

	Engagement_Score	Experience_Score
0	0.045570	0.925848
1	0.073521	0.591189

LinearRegression ⓘ ?

LinearRegression()

```
y_pred = regression_model.predict(X_test)

mse = mean_squared_error(y_test, y_pred)
rmse = mse ** 0.5
r2 = r2_score(y_test, y_pred)

print("RMSE:", rmse)
print("R² Score:", r2)
```

RMSE: 0.035144720451971236
R² Score: 0.8193658145186125

EXPERIENCE SCORE

THE SELECTED EXPERIENCE METRICS WERE AGGREGATED AT CUSTOMER LEVEL. THE METRICS WERE NORMALIZED SO THAT THEY COULD BE COMBINED INTO A COMMON SCORE.

EXPERIENCE SEGMENTATION

CUSTOMERS CAN THEN BE GROUPED BASED ON NETWORK EXPERIENCE USING CLUSTERING.

- INTERPRETATION**
- THE SEGMENTATION HELPS IDENTIFY:
- CUSTOMERS EXPERIENCING BETTER NETWORK PERFORMANCE
 - CUSTOMERS WITH MODERATE EXPERIENCE
 - CUSTOMERS POTENTIALLY EXPERIENCING POORER NETWORK CONDITIONS

CUSTOMER SATISFACTION

SATISFACTION ANALYSIS

CUSTOMER SATISFACTION
IS DERIVED FROM THE
RELATIONSHIP BETWEEN

ENGAGEMENT

- HIGHLY ENGAGED
- EXPERIENCING GOOD
NETWORK PERFORMANCE

EXPERIENCE

- POTENTIALLY VALUABLE FOR
RETENTION
- SHOWING WEAKER EXPERIENCE
DESPITE HIGH USAGE

MACHINE LEARNING / REGRESSION

PREDICTIVE ANALYSIS

A REGRESSION MODEL WAS PLANNED/IMPLEMENTED TO INVESTIGATE THE RELATIONSHIP BETWEEN CUSTOMER SATISFACTION AND THE UNDERLYING ENGAGEMENT/EXPERIENCE VARIABLES.

POTENTIAL INPUTS

- Engagement score
- Experience score
- Session behaviour
- Data consumption
- Network performance metrics

OBJECTIVE

Determine which measurable customer/network characteristics are most strongly associated with the satisfaction measure.

EVALUATION

- R^2
- MAE
- RMSE

TECHNICAL ARCHITECTURE

PROJECT ARCHITECTURE

TECHNOLOGIES

- PYTHON
- PANDAS
- NUMPY
- SCIKIT-LEARN
- MYSQL
- MLFLOW
- POWER BI /
DASHBOARD
- DOCKER
- GIT / CI-CD

Technologies Used

Programming & Analysis

- Python
- Pandas
- NumPy
- Scikit-learn
- SciPy

Visualization

- Matplotlib
- Seaborn
- Plotly
- Streamlit

Machine Learning

- K-Means Clustering
- Linear Regression

Database

- MySQL
- mysql-connector-python

MLOps & Deployment

- MLflow
- FastAPI
- Docker

DASHBOARD

BUISNESS INTELLIGENCE LAYER

DASHBOARD KPIS

- TOTAL CUSTOMERS
- AVERAGE SESSIONS/CUSTOMER
- AVERAGE ENGAGEMENT
- AVERAGE EXPERIENCE
- SATISFACTION DISTRIBUTION
- CUSTOMER SEGMENTS
- HANDSET DISTRIBUTION
- NETWORK PERFORMANCE



KEY FINDINGS

DATASET

- 150,001 INITIAL RECORDS
- 148,506 RECORDS AFTER CLEANING
- 106,856 UNIQUE CUSTOMERS

ENGAGEMENT

- AVERAGE CUSTOMER HAS APPROXIMATELY 1.39 SESSIONS
- CUSTOMER ENGAGEMENT VARIES CONSIDERABLY
- ENGAGEMENT SCORING ALLOWS CUSTOMERS TO BE RANKED AND SEGMENTED

DATA QUALITY

- SIGNIFICANT MISSINGNESS EXISTS IN SELECTED NETWORK VARIABLES
- SEVERAL FEATURES SHOW VERY HIGH CORRELATION/REDUNDANCY
- OUTLIERS ARE PRESENT IN THROUGHPUT-RELATED VARIABLES

EXPERIENCE

NETWORK QUALITY CAN BE CHARACTERIZED THROUGH:

- TCP RETRANSMISSION
- RTT
- THROUGHPUT

CONCLUSION

WHAT THE ANALYSIS ENABLES

- Better understanding of customer behaviour
- Identification of engagement patterns
- Detection of network experience differences
- Customer segmentation
- Data-driven satisfaction analysis
- Potential support for customer retention and acquisition strategies

FUTURE SCOPE

- Improve predictive satisfaction modeling
- Deploy real-time customer scoring
- Integrate additional customer/business data
- Automate model retraining
- Monitor model performance
- Expand dashboard capabilities
- Deploy the complete pipeline using Docker/CI-CD